Table of Contents

Section 1: Academic Curriculum Structure

Bachelor of Computer Applications (BCA) - Semester 7

COURSE CODE	COURSE NAME	CREDITS	WEEKLY HOURS
US07MABCA01	Basics of Generative AI	4	4
US07MABCA02	Fundamentals of Cyber Security	4	4
US07MABCA03	Practical based on US07MABCA01	4	8
US07MABCA04	Scripting with PHP - II	2	2
US07MABCA05	Practical based on US07MABCA04	2	4
US07SEBCA06	On Job Training	6	12
Total		**22**	**34**

Bachelor of Computer Applications (BCA) - Semester 8

COURSE CODE	COURSE NAME	CREDITS	WEEKLY HOURS
US08MABCA01	Basics of Machine Learning	4	4
US08MABCA02	Cloud Computing	4	4
US08MABCA03	Practical based on US08MABCA01	4	8
US08MABCA04	Data Analytics Using R Language	2	2
US08MABCA05	Practical based on US08MABCA04	2	4
US08SEBCA06	On Job Training	6	12
Total		**22**	**34**

Section 2: Generative AI (US07MABCA01) Lab Syllabus

Below is the complete official laboratory syllabus list for Generative AI (US07MABCA01). It includes 27 core algorithmic and NLP-focused practical exercises, starting from basic string-level text manipulations, progressing through advanced sequence classification pipelines, and concluding with state-of-the-art transformer modeling. Additionally, it includes three self-development (Self Dev) projects intended for deeper autonomous study.

NO.	PRACTICAL TASK DESCRIPTION / PROBLEM STATEMENT
1	Write a Python program to sort the words of a given string in alphabetical order.
2	Write a Python program to search for a specified word in a given string.
3	Write a Python program to format and display variables within a string using string formatting techniques.
4	Write a Python program to find a specified word in a string and replace it with another word.
5	Write a Python program to remove all occurrences of a specified word from a given string.
6	Write a Python program to count the total number of words in a given string.
7	Write a Python program to remove all punctuation characters from a given string.
8	Write a Python program to split a given string into tokens using a specified delimiter.
9	Write a Python program to convert a given string into uppercase, lowercase, title case, capitalize case, and swap case.
10	Write a Python program to split a string into individual words and then join them using a specified separator.
11	Write a Python program to summarize a given English paragraph into a concise summary using a pre-trained model.
12	Write a Python program to translate a given English sentence into Hindi using a pre-trained model.
13	Write a Python program to paraphrase a given paragraph while preserving its original meaning using a pre-trained model.
14	Write a Python program to simplify a complex paragraph into easy-to-understand English using a pre-trained model.

15	Write a Python program to identify and correct grammatical errors in a given English paragraph using a pre-trained model.
16	Write a Python program to expand a short paragraph into a more detailed paragraph using a pre-trained model.
17	Write a Python program to generate a short paragraph on a given topic using a pre-trained language model.
18	Write a Python program to complete an incomplete English sentence based on the given text using a pre-trained model.
19	Write a Python program to answer questions based on a given English passage using a pre-trained question-answering model.
20	Write a Python program to classify the sentiment of a given English movie review as Positive, Negative, or Neutral using a pre-trained model.
21	Write a Python program to classify a given news article into one of the categories: Sports, Business, Technology, or Politics using a pre-trained model.
22	Write a Python program to identify and extract the names of persons, organizations, locations, and dates from a given English paragraph using a pre-trained model.
23	Write a Python program to extract the five most important keywords from a given English document using a pre-trained model.
24	Write a Python program to rewrite a given English paragraph in a formal writing style using a pre-trained model.
25	Write a Python program to identify the primary topic of a given English document from the categories: Education, Health, Technology, Business, or Sports using a pre-trained model.
26	Write a Python program to classify a given email message as Spam or Not Spam using a pre-trained model.
27	Write a Python program to detect the language of a given text from English, Hindi, Gujarati, French, German, or Spanish using a pre-trained model.
28	Self Dev - 1 (Handwritten Syllabus Extension)
29	Self Dev - 2 (Handwritten Syllabus Extension)
30	Self Dev - 3 (Handwritten Syllabus Extension)

Section 3: Practical Program Solutions & Lab Implementations

This section presents complete, verified Python program solutions matching every practical task definition in the syllabus. The first 9 programs focus on core Python string programming techniques, string functions, and basic structural transformations. Starting with Program 11, the implementations show the usage of Hugging Face's pipeline framework and the transformers package to solve complex natural language processing tasks using pre-trained NLP models.

Program 1: Alphabetical Word Sorting

Statement: *Write a Python program to sort the words of a given string in alphabetical order.*

```
text = "banana apple mango orange"
words = text.split()
words.sort()
print(words)
print(" ".join(words))
```

Program 2: String Word Search

Statement: *Write a Python program to search for a specified word in a given string.*

```
text = "Python is easy to learn"
word = "easy"
if word in text:
    print("Found")
else:
    print("Not Found")
```

Program 3: Variable String Formatting

Statement: *Write a Python program to format and display variables within a string using string formatting techniques.*

```
name = "Nehal"
age = 51
print(f"My name is {name} and I am {age} years old.")
print("My name is {} and I am {} years old.".format(name, age))
```

Program 4: Substring Search and Replacement

Statement: *Write a Python program to find a specified word in a string and replace it with another word.*

```
text = "I like Java."
new_text = text.replace("Java", "Python")
print(new_text)
```

Program 5: String Content Filtering (Removing Occurrences)

Statement: *Write a Python program to remove all occurrences of a specified word from a given string.*

```
text = "Python is very very easy"
words = text.split()
filtered = [word for word in words if word != "very"]
print(" ".join(filtered))
```

Program 6: Word Frequency Counting

Statement: *Write a Python program to count the total number of words in a given string.*

```
text = "Python is easy to learn"
count = len(text.split())
print(count)
```

Program 7: Punctuation Character Stripping

Statement: *Write a Python program to remove all punctuation characters from a given string.*

```
import string
text = "Hello, World! Welcome."
result = text.translate(str.maketrans('', '', string.punctuation))
print(result)
```

Program 8: String Tokenization by Delimiter

Statement: *Write a Python program to split a given string into tokens using a specified delimiter.*

```
text = "Apple,Banana,Mango,Orange"
tokens = text.split(",")
print(tokens)
```

Program 9: String Letter Case Transformations

Statement: *Write a Python program to convert a given string into uppercase, lowercase, title case, capitalize case, and swap case.*

```
text = "Python Programming"
print(text.upper())
print(text.lower())
print(text.title())
print(text.capitalize())
print(text.swapcase())
```

Program 10: Splitting and Joining with Custom Separator

Statement: *Write a Python program to split a string into individual words and then join them using a specified separator.*

```
text = "Python is easy"
words = text.split()
print(words)
joined = "-".join(words)
print(joined)
```

Program 11: Hugging Face Text Summarization

Statement: *Write a Python program to summarize a given English paragraph into a concise summary using a pre-trained model.*

```
# pip install transformers torch sentencepiece
from transformers import pipeline
```

```
summarizer = pipeline("summarization", model="sshleifer/distilbart-cnn-12-6")
text = "Your long text here..."
result = summarizer(text, max_length=60, min_length=20, do_sample=False)
print(result[0]["summary_text"])
```

Program 12: Machine Translation Pipeline

Statement: *Write a Python program to translate a given English sentence into Hindi using a pre-trained model.*

```
# pip install transformers torch sentencepiece
from transformers import pipeline

translator = pipeline("translation", model="Helsinki-NLP/opus-mt-en-fr")
text = "Artificial Intelligence is changing the world."
result = translator(text)
print(result[0]["translation_text"])
```

Program 13: Generative Paragraph Paraphrasing

Statement: *Write a Python program to paraphrase a given paragraph while preserving its original meaning using a pre-trained model.*

```
# pip install transformers torch sentencepiece
from transformers import pipeline

paraphraser = pipeline("text2text-generation",
model="humarin/chatgpt_paraphraser_on_T5_base")
text = "Python is an easy programming language."
result = paraphraser(text)
print(result[0]["generated_text"])
```

Program 14: Sentence Simplification

Statement: *Write a Python program to simplify a complex paragraph into easy-to-understand English using a pre-trained model.*

```
# pip install transformers torch sentencepiece
from transformers import pipeline

simplifier = pipeline("text2text-generation", model="t5-small")
text = "simplify: The physician administered medication."
result = simplifier(text)
print(result[0]["generated_text"])
```

Program 15: Grammatical Error Correction

Statement: *Write a Python program to identify and correct grammatical errors in a given English paragraph using a pre-trained model.*

```
# pip install transformers torch sentencepiece
from transformers import pipeline

corrector = pipeline("text2text-generation", model="vennify/t5-base-grammar-correction")
text = "grammar: He go to school yesterday."
result = corrector(text)
print(result[0]["generated_text"])
```

Program 16: Generative Paragraph Expansion

Statement: *Write a Python program to expand a short paragraph into a more detailed paragraph using a pre-trained model.*

```
# pip install transformers torch sentencepiece
from transformers import pipeline

generator = pipeline("text2text-generation", model="google/flan-t5-base")
text = "Expand: AI helps doctors."
result = generator(text)
print(result[0]["generated_text"])
```

Program 17: Prompt-Based Paragraph Generation

Statement: *Write a Python program to generate a short paragraph on a given topic using a pre-trained language model.*

```
# pip install transformers torch
from transformers import pipeline

generator = pipeline("text-generation", model="gpt2")
result = generator("Artificial Intelligence", max_length=80)
print(result[0]["generated_text"])
```

Program 18: Autoregressive Sentence Completion

Statement: *Write a Python program to complete an incomplete English sentence based on the given text using a pre-trained model.*

```
# pip install transformers torch
from transformers import pipeline

generator = pipeline("text-generation", model="gpt2")
result = generator("Python is", max_length=50)
print(result[0]["generated_text"])
```

Program 19: Extractive Question Answering

Statement: *Write a Python program to answer questions based on a given English passage using a pre-trained question-answering model.*

```
# pip install transformers torch
from transformers import pipeline

qa = pipeline("question-answering")
result = qa(
    question="What is the capital of France?",
    context="Paris is the capital of France."
)
print(result["answer"])
```

Program 20: Sentiment Analysis Classifier

Statement: *Write a Python program to classify the sentiment of a given English movie review as Positive, Negative, or Neutral using a pre-trained model.*

```
# pip install transformers torch
from transformers import pipeline

classifier = pipeline("sentiment-analysis")
result = classifier("I love Python.")
print(result)
```

Program 21: Sequence Classification & Fine-Tuning

Statement: *Write a Python program to classify a given news article into one of the categories: Sports, Business, Technology, or Politics using a pre-trained model.*

```
# pip install transformers torch
from transformers import pipeline

classifier = pipeline("text-classification", model="distilbert-base-uncased-finetuned-sst-2-english")
result = classifier("This product is excellent.")
print(result)
```

Program 22: Named Entity Recognition (NER)

Statement: *Write a Python program to identify and extract the names of persons, organizations, locations, and dates from a given English paragraph using a pre-trained model.*

```
# pip install transformers torch
from transformers import pipeline

ner = pipeline("ner", grouped_entities=True)
result = ner("Elon Musk founded SpaceX in the United States.")
print(result)
```

Program 23: Unsupervised Keyword Extraction

Statement: *Write a Python program to extract the five most important keywords from a given English document using a pre-trained model.*

```
# pip install keybert sentence-transformers
from keybert import KeyBERT

kw_model = KeyBERT()
text = "Artificial Intelligence and Machine Learning are transforming healthcare."
keywords = kw_model.extract_keywords(text)
print(keywords)
```

Program 24: Formal Text Style Rewriting

Statement: *Write a Python program to rewrite a given English paragraph in a formal writing style using a pre-trained model.*

```
# pip install transformers torch sentencepiece
from transformers import pipeline

rewriter = pipeline("text2text-generation", model="google/flan-t5-base")
text = "Rewrite formally: Send me the report."
result = rewriter(text)
print(result[0]["generated_text"])
```

Program 25: Zero-Shot Topic Classification

Statement: *Write a Python program to identify the primary topic of a given English document from the categories: Education, Health, Technology, Business, or Sports using a pre-trained model.*

```
# pip install transformers torch
from transformers import pipeline

classifier = pipeline("zero-shot-classification")
result = classifier(
    "Python is widely used in Artificial Intelligence.",
    candidate_labels=["Technology", "Sports", "Politics", "Health"]
)
print(result)
```

Program 26: SMS Spam Detection Classification

Statement: *Write a Python program to classify a given email message as Spam or Not Spam using a pre-trained model.*

```
# pip install transformers torch
from transformers import pipeline

classifier = pipeline("text-classification", model="mrm8488/bert-tiny-finetuned-sms-spam-detection")
result = classifier("Congratulations! You have won a free iPhone.")
print(result)
```

Program 27: Multilingual Language Detection

Statement: *Write a Python program to detect the language of a given text from English, Hindi, Gujarati, French, German, or Spanish using a pre-trained model.*

```
# pip install transformers torch sentencepiece
from transformers import pipeline

detector = pipeline("text-classification", model="papluca/xlm-roberta-base-language-detection")
result = detector("Bonjour tout le monde")
print(result)
```